Hadamard's Formula for Eigenvalue Variations

Liyan Gong

Partial Differential Equations; Prof. Vlad Vicol

Homework 3, Problem 15 (Evans, Section 6.6); original coursework excerpt

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Problem 15.

(Eigenvalues and domain variations) Consider a family of smooth, bounded domains $U(\tau) \subset \mathbb{R}^n$ that depend smoothly upon the parameter $\tau \in \mathbb{R}$. As τ changes, each point on $\partial U(\tau)$ moves with velocity $\mathbf{v}$.

For each τ, we consider eigenvalues $\lambda = \lambda(\tau)$ and corresponding eigenfunctions $w = w(x, \tau)$:

$$\begin{cases} -\Delta w = \lambda w & \text{in } U(\tau) \\ w = 0 & \text{on } \partial U(\tau), \end{cases}$$

normalized so that $\|w\|_{L^2(U(\tau))} = 1$. Suppose that λ and w are smooth functions of τ and x.

Prove *Hadamard's variational formula*

$$\dot{\lambda} = -\int_{\partial U(\tau)} \left| \frac{\partial w}{\partial \nu} \right|^2 \mathbf{v} \cdot \nu \, dS,$$

where $\dot{} = \frac{d}{d\tau}$ and $\mathbf{v} \cdot \nu$ is the normal velocity of $\partial U(\tau)$.

(Hint: Use the calculus formula from §C.4.)

Proof:

Step 1: Differentiate the normalization condition

We are given the L^2 normalization condition:

$$\int_{U(\tau)} w^2 \, dx = 1$$

Differentiating both sides with respect to τ and applying the transport theorem (the calculus formula from the hint) for moving domains, we obtain:

$$0 = \frac{d}{d\tau} \int_{U(\tau)} w^2 \, dx = \int_{U(\tau)} \frac{\partial (w^2)}{\partial \tau} \, dx + \int_{\partial U(\tau)} w^2 (\mathbf{v} \cdot \nu) \, dS$$

$$0 = \int_{U(\tau)} 2w\dot{w} \, dx + \int_{\partial U(\tau)} w^2 (\mathbf{v} \cdot \nu) \, dS$$

Since $w = 0$ on the boundary $\partial U(\tau)$, the boundary integral vanishes. This yields the orthogonality relation:

$$\int_{U(\tau)} w\dot{w} \, dx = 0$$

Step 2: Differentiate the eigenvalue equation

In the interior of $U(\tau)$, the eigenfunction satisfies $-\Delta w = \lambda w$. Differentiating both sides with respect to τ (noting that the spatial Laplacian Δ commutes with the temporal derivative $\frac{d}{d\tau}$), we get:

$$-\Delta \dot{w} = \dot{\lambda} w + \lambda \dot{w}$$

Step 3: Differentiate the boundary condition

On the moving boundary $\partial U(\tau)$, we have $w(x(\tau), \tau) = 0$. We must take the total derivative with respect to τ using the chain rule, recognizing that the boundary points move with velocity $\frac{dx}{d\tau} = \mathbf{v}$:

$$\frac{d}{d\tau} w(x(\tau), \tau) = \nabla w \cdot \frac{dx}{d\tau} + \frac{\partial w}{\partial \tau} = \nabla w \cdot \mathbf{v} + \dot{w} = 0$$

Thus, on the boundary $\partial U(\tau)$, we have:

$$\dot{w} = -\nabla w \cdot \mathbf{v}$$

Furthermore, since $w \equiv 0$ on $\partial U(\tau)$ (the boundary is a level set of w), the spatial gradient ∇w must be parallel to the outward unit normal vector ν. Therefore, we can write $\nabla w = \frac{\partial w}{\partial \nu} \nu$. Substituting this into the equation above yields the boundary condition for $\dot{w}$:

$$\dot{w} = -\left(\frac{\partial w}{\partial \nu} \nu \right) \cdot \mathbf{v} = -\frac{\partial w}{\partial \nu} (\mathbf{v} \cdot \nu) \quad \text{on } \partial U(\tau)$$

Step 4: Isolate $\dot{\lambda}$ using integration

Multiply the differentiated equation from Step 2 by w and integrate over $U(\tau)$:

$$\int_{U(\tau)} -w \Delta \dot{w} \, dx = \int_{U(\tau)} (\dot{\lambda} w^2 + \lambda w \dot{w}) \, dx = \dot{\lambda} \int_{U(\tau)} w^2 \, dx + \lambda \int_{U(\tau)} w \dot{w} \, dx$$

Using the normalization condition ($\int_{U(\tau)} w^2 dx = 1$) and the result from Step 1 ($\int_{U(\tau)} w \dot{w} dx = 0$), the right side simplifies to just $\dot{\lambda}$:

$$\dot{\lambda} = \int_{U(\tau)} -w \Delta \dot{w} \, dx$$

Now, apply Green's second identity to the integral on the right:

$$\int_{U(\tau)} (-w \Delta \dot{w}) \, dx = \int_{U(\tau)} (-\dot{w} \Delta w) \, dx - \int_{\partial U(\tau)} \left(w \frac{\partial \dot{w}}{\partial \nu} - \dot{w} \frac{\partial w}{\partial \nu} \right) dS$$

We can simplify the terms on the right-hand side of this identity:

- Because $-\Delta w = \lambda w$, the first integral is $\int_{U(\tau)} \dot{w} (\lambda w) \, dx = \lambda \int_{U(\tau)} w \dot{w} \, dx = 0$ (from Step 1).
- Because $w = 0$ on $\partial U(\tau)$, the term $w \frac{\partial \dot{w}}{\partial \nu}$ vanishes.

Therefore, the identity reduces to:

$$\dot{\lambda} = \int_{\partial U(\tau)} \dot{w} \frac{\partial w}{\partial \nu} \, dS$$

Step 5: Final substitution

Substitute the expression for $\dot{w}$ on the boundary derived in Step 3 into the integral:

$$\dot{\lambda} = \int_{\partial U(\tau)} \left(-\frac{\partial w}{\partial \nu} (\mathbf{v} \cdot \nu) \right) \frac{\partial w}{\partial \nu} \, dS$$

$$\dot{\lambda} = -\int_{\partial U(\tau)} \left| \frac{\partial w}{\partial \nu} \right|^2 \mathbf{v} \cdot \nu \, dS$$

This is Hadamard's variational formula, and the proof is complete. ■